

Motivation Letter: AI Engineer - Generative AI Applications (RE3)

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Recruitment Panel

Barcelona Supercomputing Center - Centro Nacional de Supercomputación

Subject: Application for AI Engineer - Generative AI Applications (RE3), reference 336_26_DIR_IBD_RE3

Dear Members of the Recruitment Panel,

I am applying for the AI Engineer - Generative AI Applications position because the AI Factory's objective of turning validated AI concepts into reliable, reusable applications closely reflects the work I want to continue developing. My background combines enterprise agent systems, research robotics, computational modelling, and engineering in a safety-sensitive domain. I am currently completing an MSc in Modelling for Science and Engineering at the Universitat Autònoma de Barcelona, specialising in Data Science.

At Aily Labs, I have worked across the lifecycle by which a language model becomes a production analytical application. Prompt-defined MCP capabilities and a server package for go-to-market workflows evolved towards gated skills, where the runtime controls which tools and instructions are available for a task. A semantic layer then provides approved SQL table descriptions, business meaning, and usage guidance to an SQL agent, giving it direct but bounded access to enterprise information.

The same work now extends into evaluation and release discipline. Package-owned question sets define expected tables, skills, tools, and answer criteria; benchmark runs combine deterministic validators, traces, and LLM-as-a-judge scoring into an Agent IQ measure. I have therefore worked with failures across semantic context, routing, tool selection, SQL behaviour, service integration, and CI/CD rather than treating every poor answer as a prompt problem. These experiences are directly relevant to context management, tool calling, observability, evaluation, and the continuous improvement of reusable AI services before or alongside model post-training.

At IIIA-CSIC, my JAE Intro ICU 2025 scholarship and MSc thesis concern a modular language-model-based architecture for the NAO robot. The ROS 2 stack separates dialogue, grounded symbolic context, structured planning, deterministic action dispatch, execution feedback, and user reporting. My contribution has included cross-component contracts, planner and skill interfaces, failure handling, trace capture, and a validation methodology spanning dialogue, knowledge-base queries, simple and composite actions, and controlled recovery. This work has made reliability concrete: an application should expose where a decision came from, what capability was invoked, what evidence was returned, and whether the final statement is supported.

My earlier work at ABS Consulting adds an engineering perspective that I consider valuable for production AI. Supporting seismic qualification and structural assessments for nuclear power assets required disciplined modelling, standards-based reasoning, reviewable calculations, and precise technical reports. My BSc thesis also connected machine learning with physical modelling through neural-network prediction of turbulent intensity from wind-profile data.

My personal work examines the same reliability problem from the model and harness sides. iTrader is an RLVR-style trading research stack with an implemented task-conditioned PPO/GAE solver, FiLM policy and value networks, validated task specifications, decomposed and normalised reward components, local LLM proposer backends, and reproducible experiment manifests. It is intended to become a live platform, but the planned route passes through capability-profiled solver heads, grouped proposer optimisation, Market-JEPA representations, replay and shadow modes, and deterministic risk gates before execution.

The unpublished Universal Agentic Harness H0 proof provides the systems counterpart. Its abstraction-boundary model compiles a minimal task-specific view of capabilities, context, permissions, and evidence rather than exposing a global tool set. It currently implements focused contract tests, registry projection, deterministic output gates, append-only traces, and environment adapters, but is not live-integrated or externally validated. The Watson Stack adds local model serving and tool workflows through LiteLLM, llama.cpp, and a Hermes-style loop.

I recognise that this RE3 vacancy requests a longer record of production software and LLM delivery than my current title history alone demonstrates. I am nevertheless applying because the technical overlap is substantial and my trajectory has been unusually concentrated across the required boundaries: Python AI applications, agent and tool interfaces, structured context, evaluation, observability, documentation, and multidisciplinary delivery. I would bring high ownership, careful systems thinking, and the humility to treat cloud deployment, large-scale RAG, security, and LLMOps depth as areas for continued growth rather than overstating them.

BSC's combination of scientific infrastructure, public-interest AI, and cross-domain engineering is exactly the environment in which I want to build. I would welcome the opportunity to discuss how my experience could contribute to trustworthy generative AI services and reusable AI Factory components.

Yours sincerely,

Juan David Bendek Williamson

References

1. **Roger Montane**, Mid Data Scientist, Aily Labs

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